بسمه تعالی

Example: $\underline{y} = A\underline{\theta}^* + \underline{z}$ $\qquad z \sim \mathcal{N}(\underline{0}, \sigma^2 I)$

Assumption: A satisfies $RIP(2s, \gamma_{2s})$, $\|\theta^*\|_0 = s$.

Obs. $\underline{y}$, Goal: estimate θ^*, call the estimator $\hat{\theta}$.

$$\inf_{\hat{\theta}} \sup_{\theta^* \in \Theta} \mathbb{E}_{\underline{z}} \{ \| \hat{\theta} - \theta^* \|_2^2 \} \geq \omega\left(\frac{s \log d/s}{n} \right)$$

$$\Theta = \{\theta \mid \theta \in \mathbb{R}^d, \|\theta\|_0 = s, \|\theta\|_2 \leq 1\}$$

Proof: Step 1- Construct $\{\theta^{(1)}, \dots, \theta^{(M)}\}$ that is a 2δ-sep. set.

$\theta_i^{(k)} \in \{0,1\}$ $\qquad$ Subgoal: Construct "maximum number of $\theta^{(i)}$'s, i.e. M, s.t.

$$\|\theta^{(i)} - \theta^{(j)}\|_1 > s/2 .$$

$M = 2$

$$\theta^{(1)} = (\overbrace{1, 1, 1, \dots, 1}^{s}, \overbrace{0 \dots 0}^{d-s})$$

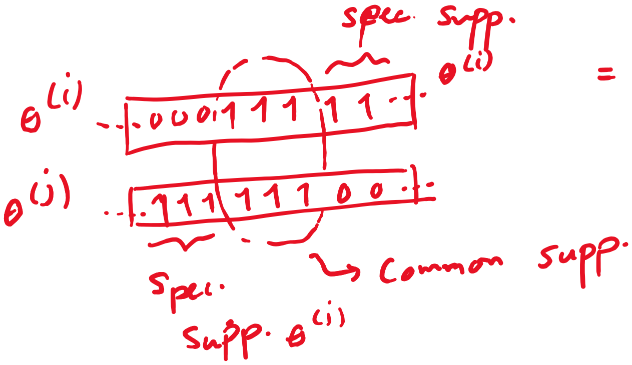
$$\theta^{(2)} = (\overbrace{0, \dots 0}^{s}, \overbrace{1, 1, \dots 1}^{s}, \overbrace{0 \dots 0}^{d-2s})$$

Randomized Construction: Consider all $\tilde{\theta}^{(i)}$'s that are s-sparse & choose M of them uniform at random. $\rightarrow$ $\theta^{(1)} \dots \theta^{(M)}$ are R.V.'s that show the chosen ones.

$$1 - D = \mathbb{P}\Big(\forall_{\substack{i,j \\ i \neq j \\ 1 \leq i,j \leq M}} \ \|\theta^{(i)} - \theta^{(j)}\|_1 \geqslant s/2 \Big) = ?$$

Consider a fixed pair of (i,j):

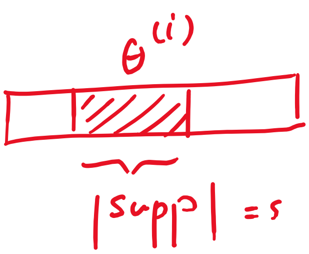
$$\mathbb{P}\Big(\underbrace{\|\theta^{(i)} - \theta^{(j)}\|_1} < s/2 \Big)$$



$$= 2\big(s - \underbrace{\langle \theta^{(i)}, \theta^{(j)} \rangle}_{\substack{\text{size of common} \\ \text{support of } \theta^{(i)} \,\&\, \theta^{(j)}}} \big)$$

$$B := \rightsquigarrow \mathbb{P}\Big(2\big(s - \langle \theta^{(i)}, \theta^{(j)} \rangle\big) < s/2 \Big)$$

$$= \mathbb{P}\Big(\underbrace{\langle \theta^{(i)}, \theta^{(j)} \rangle > 3s/4} \Big)$$



$$\langle \theta^{(i)}, \theta^{(j)} \rangle = \lceil 3s/4 \rceil$$

or $$\langle \theta^{(i)}, \theta^{(j)} \rangle = \lceil 3s/4 \rceil + 1$$

$$\vdots$$

or

$$\langle \theta^{(i)}, \theta^{(j)} \rangle = s$$

$$\mathbb{P}\big(\langle \theta^{(i)}, \theta^{(j)} \rangle = k \big) = \binom{s}{k}\binom{d-s}{s-k} \Big/ \binom{d}{s}$$

$$B = \sum_{k = 3s/4 + 1}^{s} \binom{s}{k}\binom{d-s}{s-k} \Big/ \binom{d}{s}$$

$s < d/2$

$\frac{d-s}{2} > \frac{s}{4} - 1$

$$B \leq \frac{s}{4} \cdot \binom{s}{3s/4+1} \binom{d-s}{s/4-1} \Big/ \binom{d}{s}$$

$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} \leq \left(\frac{en}{k}\right)^k$$

$$B \leq \frac{s}{4} \left[\frac{e\,s}{3s/4}\right]^{3s/4+1} \left[\frac{e(d-s)}{s/4-1}\right]^{s/4-1} \cdot \left(\frac{d}{s}\right)^{-s}$$

$$\leq s/4 \, \Big(\underbrace{\frac{4e}{3}}_{c}\Big)^{3s/4+1} \Big[\underbrace{4e}_{c'}\left(\frac{d-s}{s-4}\right)\Big]^{s/4-1} \left(\frac{d}{s}\right)^{-s}$$

$$\leq \underbrace{\frac{s}{4} \cdot (c')^{s} \left(\frac{d-s}{s-4}\right)^{s/4-1} \left(\frac{d}{s}\right)^{-s}}_{E}$$

$$\Rightarrow \quad D \leq \binom{M}{2} \cdot E \leq \frac{M^2}{2} \cdot E$$

$$\log D \leq 2 \log M + \log s/4 + s \log c' + (s/4 - 1) \log \frac{d-s}{s-4}$$
$$- s \log d/s + c''$$

$$\log M \geq \log D - \log s/4 - s \log c' - (s/4 - 1) \log \frac{d-s}{s-4} + s \log d/s$$
$$- c''$$

$\exists\, c$

$$\geq c\, s \log d/s$$

If for some event K, $\mathbb{P}(K) > 0$, there exists an outcome $K(\xi)$ s.t. $K(\xi) = \text{True}$.

$\Rightarrow$ There exists $\{\theta^{(1)}, \dots, \theta^{(M)}\}$, with $\log M = \Omega(s \log d/s)$

s.t. $\forall i,j \; \|\theta^{(i)} - \theta^{(j)}\|_1 > s/2$.